Induction

Notes 1/27

Claim: $0+1+\dots+n = \sum_{i=0}^{n} i = \frac{n(n+1)}{2}$

Proof with axiom 4: Let S be the subset of $\mathbb{N}$ for which the claim holds. I.E, $S = \{n \in \mathbb{N} : \sum_{i=0}^{n} = \frac{n(n+1)}{2}\}$

BASE CASE: To see $\emptyset \in S$, compute $\emptyset = \sum_{i=\emptyset}^{0} i$ and $\frac{0(0+1)}{2} = \emptyset$

Thus $\sum_{i=\emptyset}^{0} i = \frac{0(0+1)}{2}$, and $\emptyset \in S$

Induction STEP: Suppose $n \in S$. Thus $\sum_{i=\emptyset}^{n} i = \frac{n(n+1)}{2}$. We compute

$$\sum_{i=\emptyset}^{\sigma(n)} i = \sum_{i=0}^{n} + \sigma(n) = \sum_{i=0}^{n} i + n+1 = \frac{n(n+1)}{2} + n+1$$

$$= \frac{n(n+1)}{2} + \frac{2(n+1)}{2} = \frac{(n+2)(n+1)}{2}$$ Thus $\sigma(n) \in S$

Axiom 4 implies $S = \mathbb{N}$

Proof proceeding via induction, note $\sum_{i=0}^{0} i = \frac{0(0+1)}{2}$

Further, if $\sum_{i=0}^{n} i$, then $\sum_{i=0}^{n+1} i = \frac{n(n+1)}{2} + n+1 = \frac{(n+1)(n+2)}{2}$

Our result follows.

Informal Exercise 2: $\{\}$ empty set doesn't contain $\emptyset$ and is a proper subset of n

$\{1,2,3\}$ $\sigma(n) = n+1 \geq 2$ it doesn't contain $\emptyset$

$\{4,5,6\dots\}$ Doesn't contain $\emptyset$, successors stay in the set.

S closed under successors means if $n \in S$, then $\sigma(n) \in S$

Examples

$S = \{n \in \mathbb{N} : n > 2\}$

$S = \{n \in \mathbb{N} : n \geq k\}$ for some $k \in \mathbb{N}^+$

Def. If A and B are $\mathbb{N}$ such that $\sigma(a) = b$ then we say b has a predecessor $\underline{a}$ or a is the predecessor of b

Theorem 1: The natural # $\emptyset$ has no predecessor in $\mathbb{N}$

Proof: Suppose that 0 has a predecessor $x \in \mathbb{N}$
By def of predecessor $\sigma(x) = 0$ However
this contradicts Axiom 2.
Thus $\emptyset$ has no predecessor in $\mathbb{N}$

Theorem 2: Every nonzero element of $\mathbb{N}$ has a predecessor in $\mathbb{N}$

Proof: We ~~must~~ proceed via Axiom 4
Let S be the set of natural numbers with a predecessor union $\{0\}$ I.E., $S = \{n \in \mathbb{N}$ n has a predecessor $\cup \{0\}$

S contains $\emptyset$, and is a subset of $\mathbb{N}$. Now suppose $n \in S$. We want to show $\sigma(n)$ is in S
Thus we need to show $\sigma(n)$ has a predecessor
Show that there is some $k \in \mathbb{N}$ such that $\sigma(k) = \sigma(n)$
But of course n is the predecessor of $\sigma(n)$.
Thus, $\sigma(n)$ has a predecessor in $\mathbb{N}$ and is in S